

20250603EB_MN (GCamp with Stephanie)

Project: Yi-Shuan Tseng Holt Lab

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2025年6月3日 星期二

Differentiation: BP w/ 1.5mM pyruvate.

split iPSCs into a 5 wells of 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年6月5日 星期四

T2 -> T3

*try to use accutase to detach the cell this time.

1. Wash cells with PBS
2. Add 1mL accutase and incubate the plate in 37C
3. After cells detached, take 150uL cell suspension into a well of 6-well plate. 2 wells in total with Stemflex media +ROCK1
4. Add 1mL PBS into each well with accutase and collect all the suspension from 5 wells
5. Centrifuge at 200xg for 3 min
6. Gently resuspend cells into 10mL T2
7. Plate cells into cell-repellent 10cm dish

2025年6月7日 星期六

T3 -> T4 (~14 mL for over-weekend)

2025年6月9日 星期一

T4 -> T5

2025年6月11日 星期三

T5 -> T6

2025年6月13日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年6月16日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) o/n
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine over weekend

4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

NOTE

- Use newly opened 0.25% tpsin and aliquot 20mL to add 1 vial of DNase.
- Combine 1.5mL old trypsin with 1.5mL newly prepared trypsin
- The trypsin is very strong and the cells clump and became viscous within 5 min

coating date

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	A	B
1	0.01% PLO with Na borate	2025/6/6
2	laminin	2025/6/7

cell info

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	A	B	C
1	cell	passage	well#
2	WT11	P16+15	5

Table1

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	A	B
1	run1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6

Table3								
	A	B	C	D	E	F	G	H
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells
2	0.38	99	5	1.9	A	24	237.5	8

Well1						
	1	2	3	4	5	6
A				vLH74 (20uL in total)		
B						
C						
D						

- 2025年6月18日 星期三

Change T7 -> T7
- 2025年6月19日 星期四

T7 -> BP T8
- 2025年6月20日 星期五

1/2 T8 -> T8
- 2025年6月23日 星期一

1/2 T8 -> 1/2 T8
- 2025年6月25日 星期三

Imaging by Stephanie

GCamp

Final protocol: image 2 position at the same time with 4s interval. 200ms exposure time with GFP, Dsred Dual camera